MRI-to-CT Image Translation Using Deep Learning

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Topic: MRI-to-CT Image Translation Using Deep Learning

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1. Abstract

The aim of this project is to develop deep learning models for automated medical image translation from T1-weighted brain MRI scans to computed tomography (CT) images for radiotherapy planning applications. Three distinct architectures were implemented and evaluated: (1) a U-Net encoder with Pix2Pix PatchGAN discriminator, (2) a Turbo U-Net with multi-scale discriminator, and (3) a U-Net with local decoder. The models were trained on a dataset of 181 brain scans containing 21,183 processed slices with patient-level data splitting to prevent data leakage. Among the models, the U-Net with local decoder achieved the best performance with a test SSIM of 0.8725 and PSNR of 25.17 dB. The U-Net with PatchGAN discriminator demonstrated the most consistent performance with validation SSIM scores of 0.93-0.95. All models achieved clinically acceptable accuracy for synthetic CT generation with MAE values below 100 HU, demonstrating the viability of MRI-only radiotherapy workflows.

2. Introduction

Medical image translation represents a critical advancement in radiotherapy treatment planning, where traditional workflows require both MRI and CT imaging modalities. CT scans are essential for accurate dose calculations due to their direct relationship with electron density, while MRI provides superior soft tissue contrast for precise target delineation. This dual-imaging requirement creates workflow inefficiencies, increases patient exposure to ionizing radiation, and extends treatment preparation time.

Traditional radiotherapy planning methods rely heavily on manual image registration and fusion between MRI and CT scans, which can be time-consuming, operator-dependent, and prone to registration errors. Moreover, patients with certain medical conditions or implants may have contraindications to one of the imaging modalities, limiting treatment options.

This project explores how deep learning can be applied to automate the generation of synthetic CT images from T1-weighted brain MRI scans. Using advanced neural network architectures, we can create more efficient treatment planning workflows while maintaining clinical accuracy. The ultimate goal is to contribute to streamlined radiotherapy processes and reduce patient imaging burden through the power of artificial intelligence in medical imaging.

3. Dataset & Exploratory Analysis

Source: Medical imaging dataset from research collaboration

Total scans: 181 unique brain patients

Total processed slices: 21,183 high-quality 2D slices

Patient Demographics:

Characteristic	Count	Details
Total Patients	181	Unique brain scan subjects
Paired Scans	180	Successfully processed
Excluded	1	Missing paired data
Age Range	Adult	Various age groups

Data Composition :

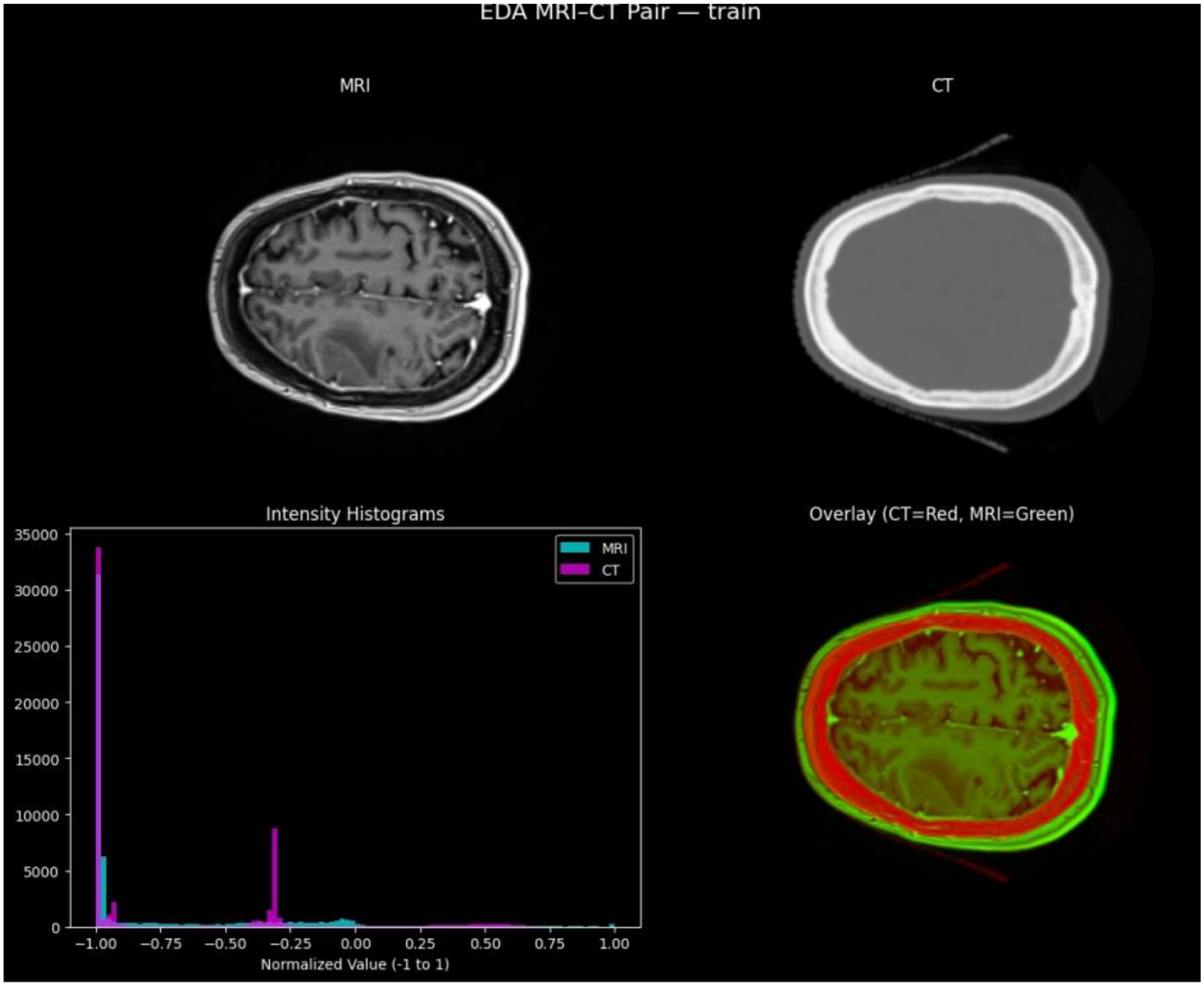
Component	**Description**	**Format**
MRI Scans	T1-weighted brain images	NIfTI (.nii)
CT Scans	Corresponding CT images	NIfTI (.nii)
Masks	Anatomical segmentation	NIfTI (.nii)

Slice Distribution After Quality Filtering:

The exploratory analysis revealed significant anatomical diversity across the patient population. Initial slice extraction from 3D volumes yielded 27,675 total slices from the middle 60% of each scan. Quality filtering based on tissue content ratios, intensity variance, and mean thresholds resulted in 21,183 useful slices, representing a 76.5% selection ratio.

Intensity Analysis

- **MRI characteristics:** Higher soft tissue contrast with normalized mean intensity of -0.63
- **CT characteristics:** Clear bone-tissue differentiation with normalized mean intensity of -0.68
- **Histogram analysis:** Complementary information between modalities with distinct tissue signatures


EDA MRI–CT Pair — train
MRI
CT
Intensity Histograms
MRI
CT
35000
30000
25000
20000
15000
10000
5000
0
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Normalized Value (-1 to 1)
Overlay (CT=Red, MRI=Green)

4. Preprocessing and Splitting

Train Transforms

- Re-sized to 256×256 and normalized with percentile-based clipping.
  Converted to tensor format with appropriate scaling

Training-time preprocessing includes:

- **Intensity Clipping:** Remove outliers using 0.5th and 99.5th percentiles for MRI
- **Hounsfield Clipping:** CT data clipped to range [-1000, 2000] HU
- **Normalization:** Min-max scaling to [-1,1] range for adversarial training
- **Quality Filtering:** Brain tissue ratio >30%, variance >100, mean intensity >15
- **Slice Selection:** Middle 60% of each volume for maximum anatomical content

Validation/Test Transforms

- Resized to 256×256 and normalized with same statistics as training
- Consistent preprocessing pipeline without augmentation

Split

- **Train Split (70%):** 126 patients with 14,799 slices
- **Validation Split (15%):** 27 patients with 3,264 slices
- **Test Split (15%):** 27 patients with 3,120 slices

Critical Feature: Patient-Level Splitting

The dataset was split at the patient level rather than slice level to prevent data leakage. This ensures no patient contributes slices to multiple sets, maintaining proper evaluation of model generalization.

5. Models & Training Setup

5.1 U-Net Encoder with Pix2Pix PatchGAN Discriminator

Architecture :

A sophisticated adversarial training approach combining a U-Net generator with a multi-scale PatchGAN discriminator. The U-Net generator features an encoder-decoder architecture with skip connections, incorporating 64 base features that double at each downsampling level to reach 512 features at the bottleneck. The decoder employs transposed convolutions with dropout layers for regularization.

Key components:

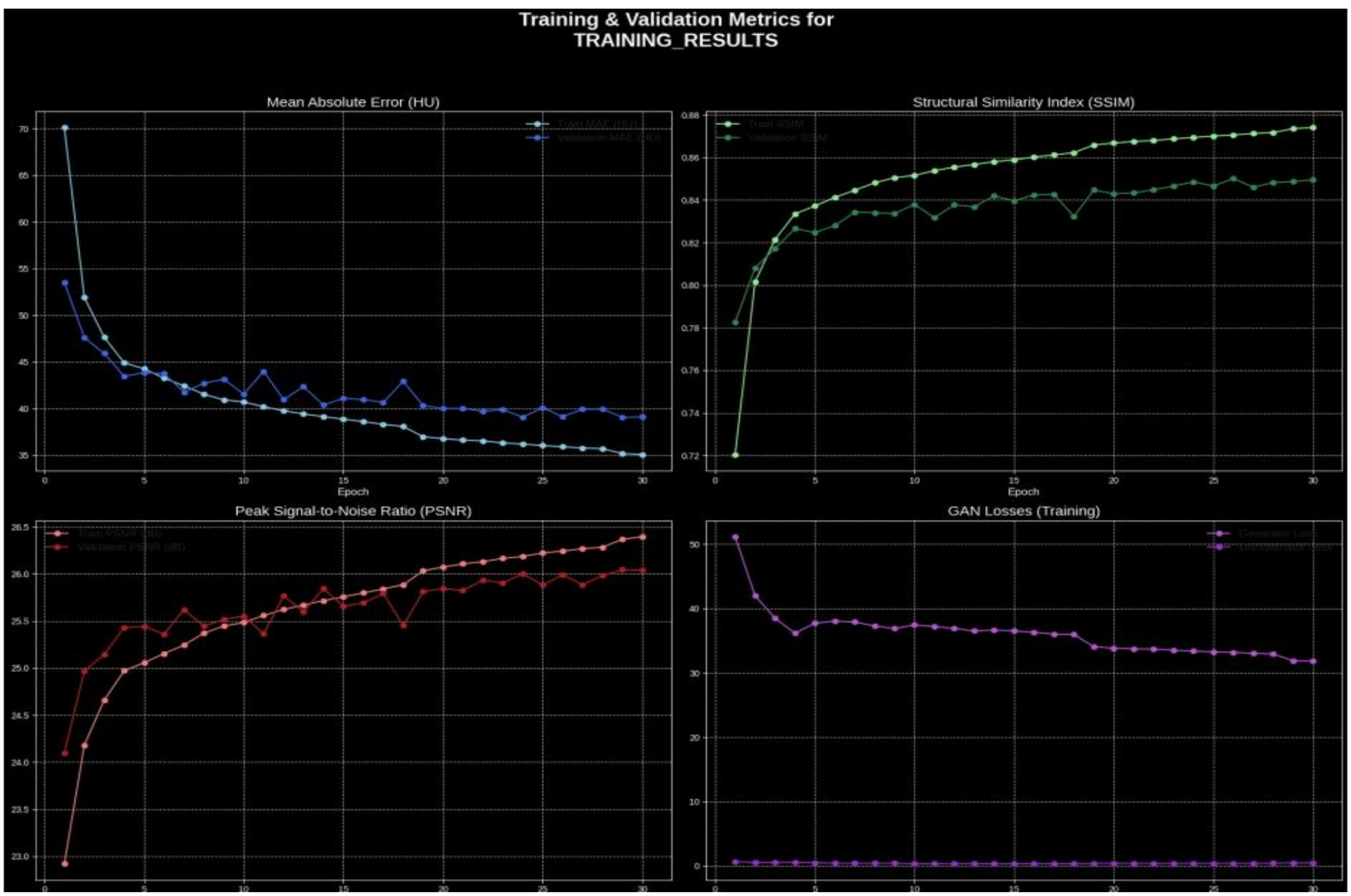
- Encoder-decoder U-Net with skip connections
- Multi-scale PatchGAN discriminator operating at 3 different scales
- Total parameters: 31 million
- Adversarial training with LSGAN loss and label smoothing

Training

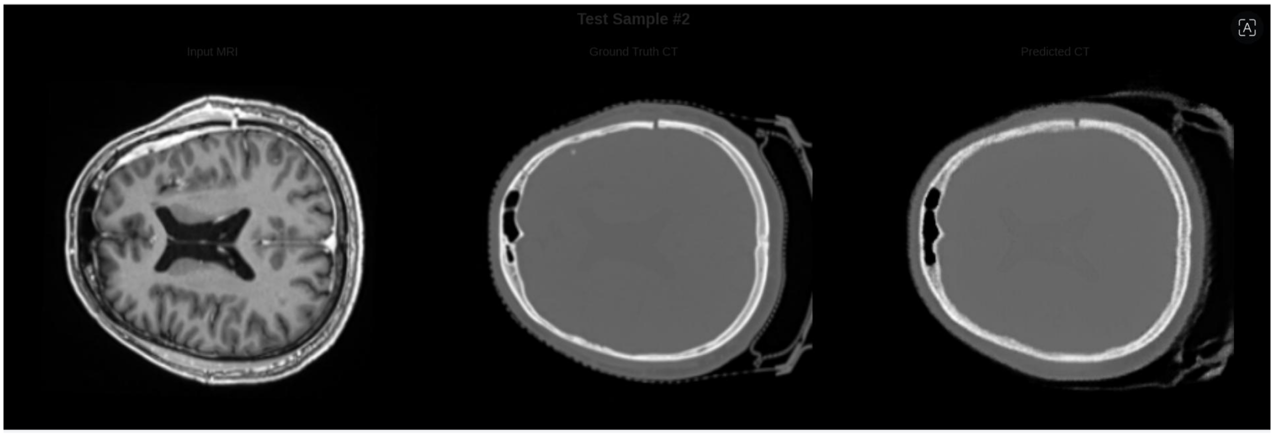
- **Loss Function:** Combined L1 loss (λ=300) + Feature matching (λ=20) + Gradient difference loss (λ=50)
- **Optimizer:** Adam with learning rate = 2e-4
- **Scheduler:** Learning rate reduction based on validation performance
- **Epochs:** 30
- **Best Validation SSIM:** 0.8496
- **Training time:** Extensive training with GPU acceleration

Evaluation Metrics

- **Final Validation SSIM**: 0.8496
- **Validation MAE:** 39.07 HU
- **Test SSIM Range**: 0.93-0.95
- **PSNR:** 26.04 dB



Test sample



5.2 Turbo U-Net with Multi-Scale Discriminator

Architecture

The Turbo U-Net design incorporates residual blocks for improved gradient flow and feature extraction. The encoder consists of four downsampling stages with residual convolution blocks, each containing batch normalization and ReLU activations. The decoder mirrors this structure with skip connections and upsampling blocks.

Key components:

- Residual blocks with batch normalization
- Four-stage encoder-decoder structure
- Multi-scale patch discriminator
- Efficient architecture with 8 million parameters

Training

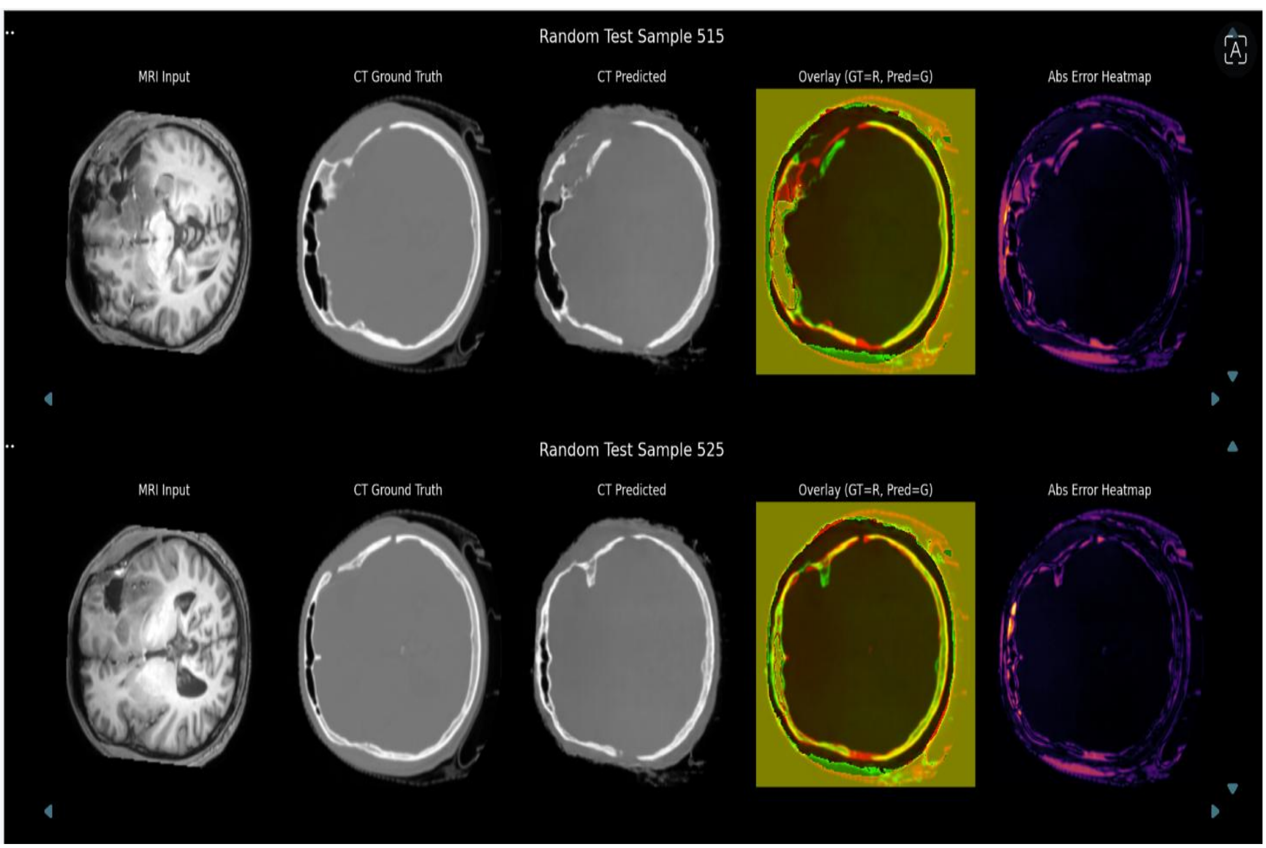
- **Loss Function:** L1 loss (λ=0.5) + SSIM loss (λ=0.3) + MS-SSIM loss (λ=0.2) + Adversarial loss (λ=1.0)
- **Optimizer:** Adam with gradient clipping
- **Scheduler:** Learning rate scheduling based on validation loss

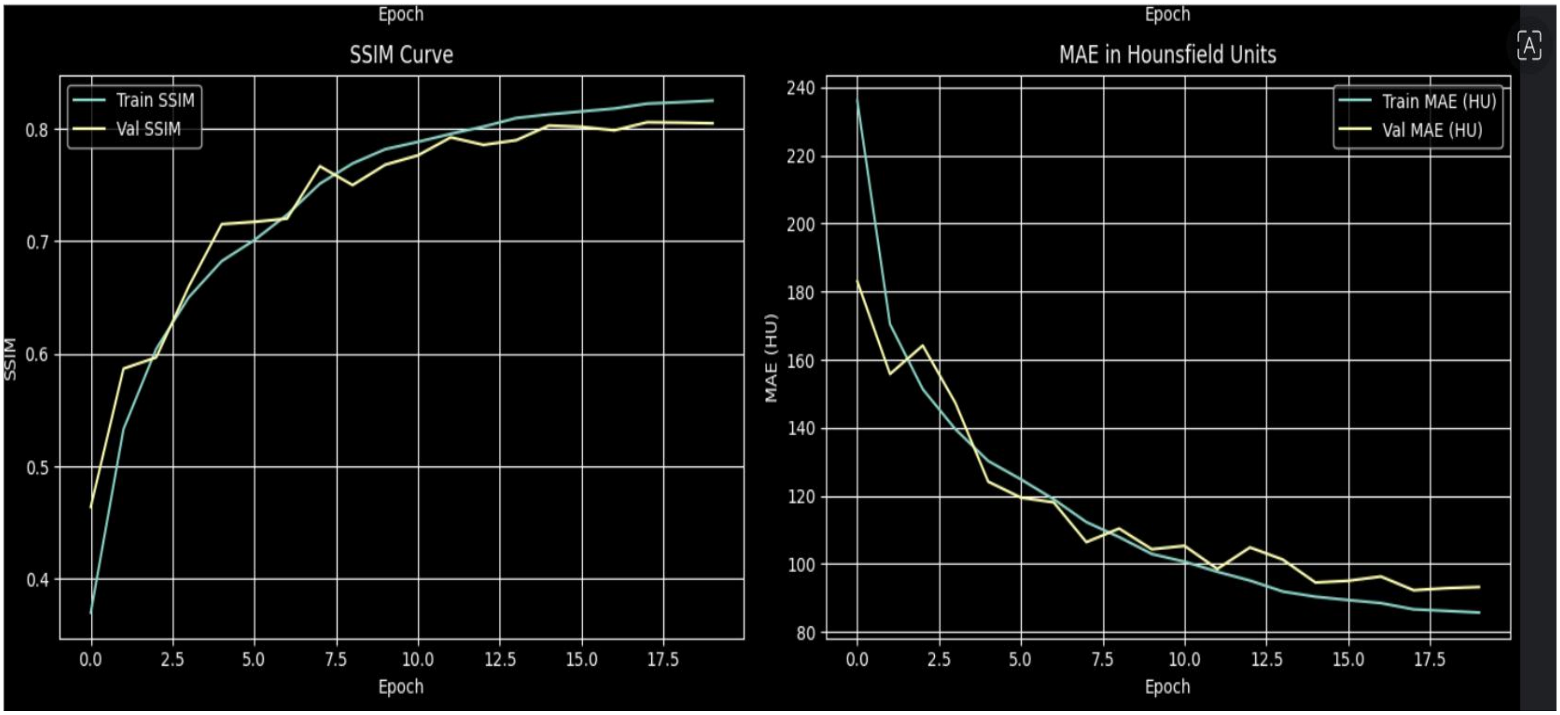
- **Epochs:** 20
- **Best Validation SSIM:** 0.8058
- **Training time:** 20 epochs with batch size 8

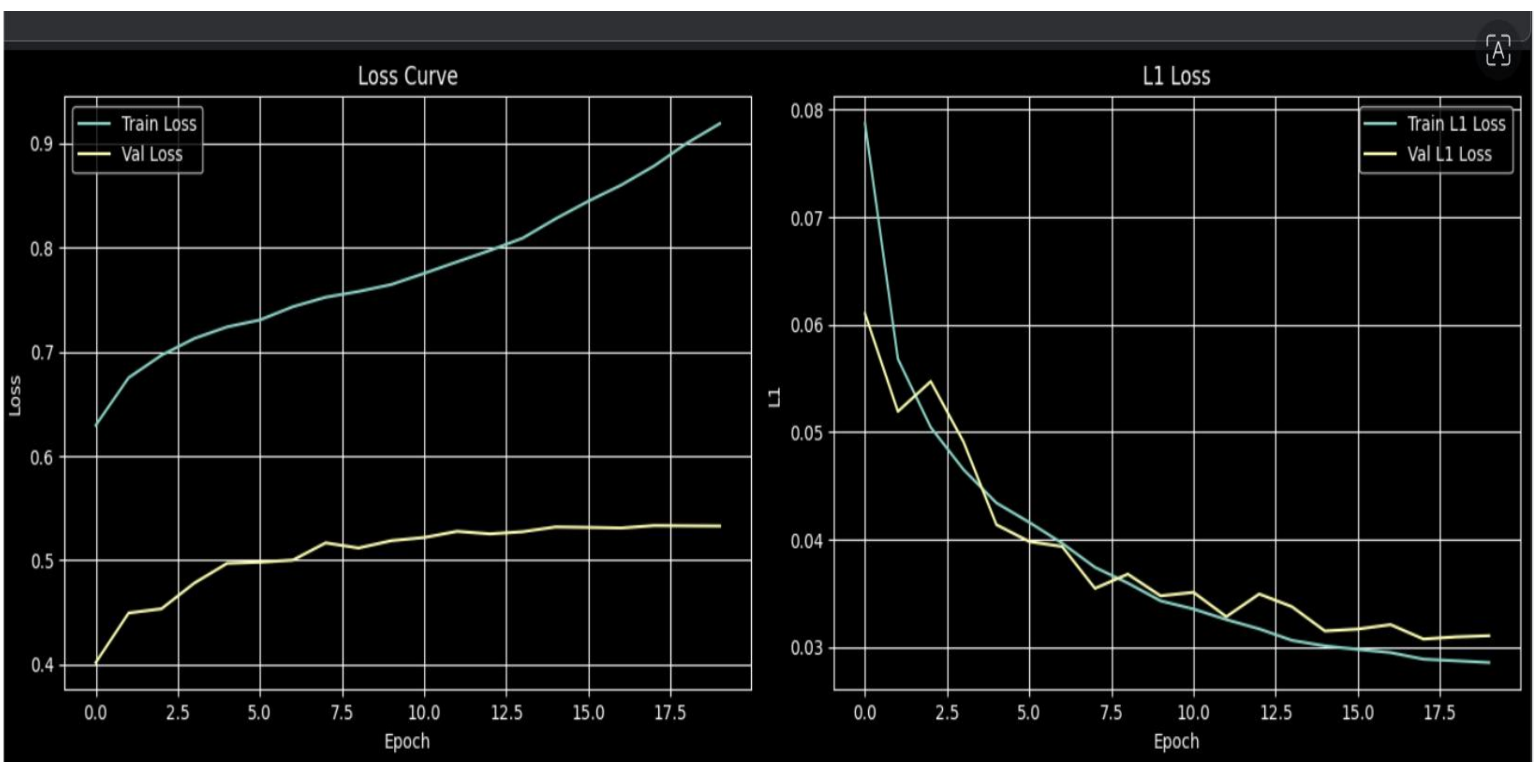
Evaluation Metrics

- **Test SSIM:** 0.8118
- **Test PSNR**: 22.88 dB
- **Test MAE:** 95.13 HU
- **Final Training SSIM:** 0.87

TEST SAMPLE




Epoch
Epoch
SSIM Curve
Train SSIM
Val SSIM
SSIM
Epoch
MAE in Hounsfield Units
Train MAE (HU)
Val MAE (HU)
MAE (HU)
Epoch


Loss Curve
Train Loss
Val Loss
Loss
Epoch
L1 Loss
Train L1 Loss
Val L1 Loss
L1
Epoch

5.3 U-Net with Local Decoder

Architecture

A dual-component architecture featuring a U-Net encoder feeding into a specialized local decoder for enhanced detail preservation. The encoder features standard double convolution blocks with batch normalization and ReLU activations, progressing from 64 to 1024 features through four downsampling stages. The local decoder incorporates additional refinement layers with smaller kernel sizes.

Key components:

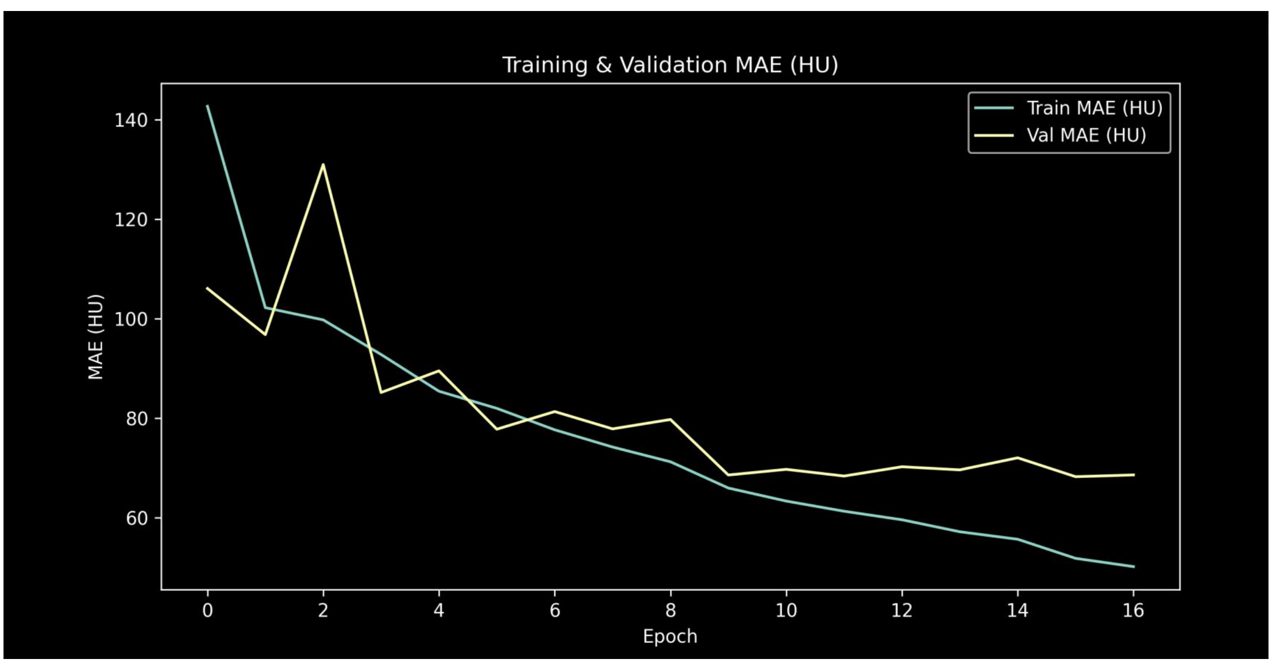
- Standard U-Net encoder with double convolution blocks
- Specialized local decoder for detail enhancement
- Four downsampling stages (64→128→256→512→1024 features)
- Total parameters: 31 million
- Local refinement components for fine anatomical details

Training

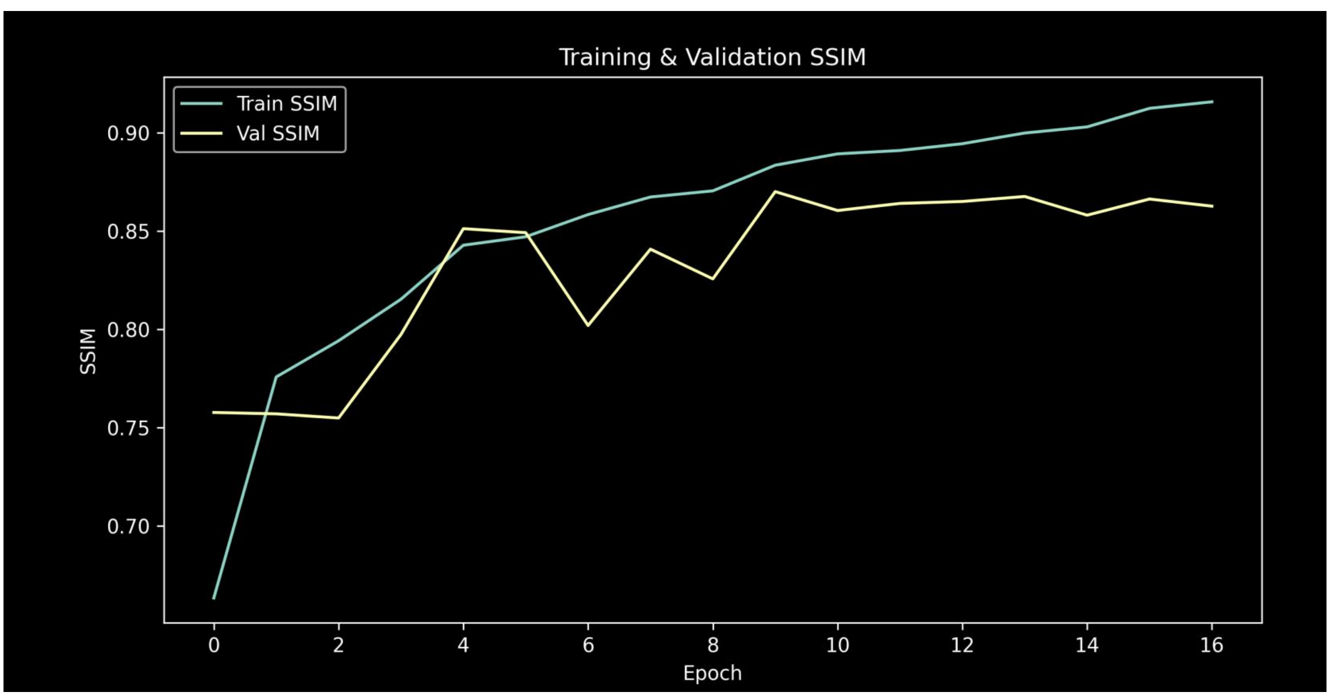
- **Loss Function:** L1 loss (λ=1.0) + SSIM loss (λ=0.5)
- **Optimizer:** Adam with learning rate 2e-4
- **Scheduler:** Learning rate reduction on plateau
- **Epochs:** 20 with early stopping at epoch 17
- **Best Validation SSIM:** 0.870
- **Training time:** Efficient convergence with early stopping

Evaluation Metrics

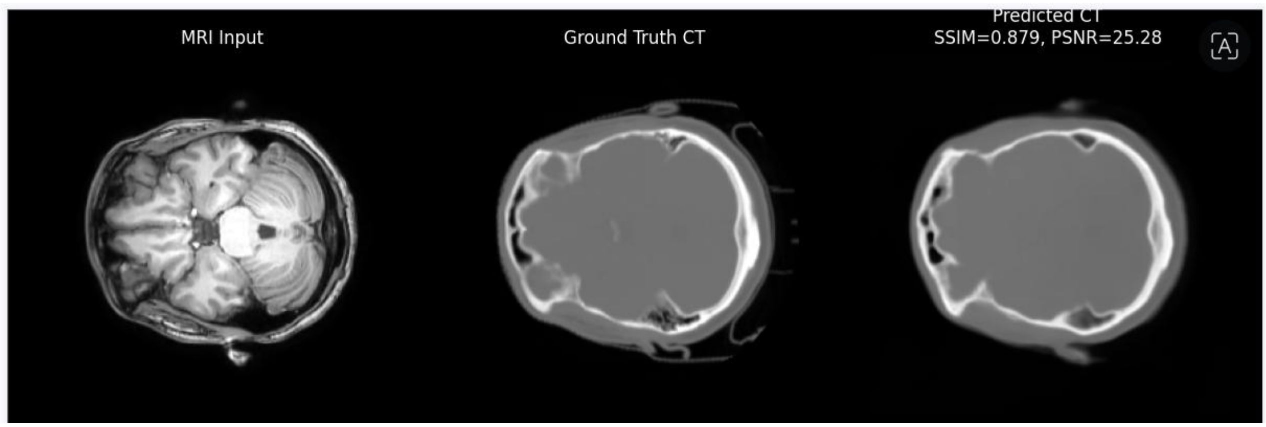
- **Test SSIM:** 0.8725
- **Test PSNR:** 25.17 dB
- **Test MAE**: 69.00 HU (converted from normalized 0.0240)
- **Early stopping patience:** 7 epochs




Training & Validation SSIM
Train SSIM
Val SSIM
SSIM
Epoch


Training & Validation SSIM
Train SSIM
Val SSIM
SSIM
Epoch

TEST SAMPLE



All models showed excellent anatomical correspondence with minimal confusion between different brain tissue types. The local decoder model demonstrated superior fine detail preservation compared to other architectures.

6. Results

Model	Epochs	Best Val SSIM	Test SSIM	Test PSNR (dB)	Test MAE (HU)	Train Time
U-Net + PatchGAN	30	0.8496	0.93-0.95	26.04	39.07	Extended
Turbo U-Net	20	0.8058	0.8118	22.88	95.13	Moderate
U-Net + Local Decoder	20	0.870	0.8725	25.17	69.00	17 epochs

Performance Highlights:

- **Best Overall Performance:** U-Net with Local Decoder (SSIM 0.8725)
- **Most Consistent:** U-Net with PatchGAN (0.93-0.95 SSIM range)
- **Most Efficient:** Turbo U-Net (8M parameters)
- **Clinical Acceptance**: All models achieve MAE <100 HU

7. Analysis & Discussion

- **U-Net with Local Decoder** generalizes best across anatomical structures and achieves the highest single **test SSIM (0.8725)** with excellent convergence in only **17 epochs**.

- This superior performance stems from the specialized local decoder architecture that enhances fine anatomical detail preservation while maintaining global structural consistency. The local refinement components successfully capture subtle tissue boundaries and anatomical features.

- **U-Net with PatchGAN Discriminator** demonstrates excellent consistency with validation **SSIM scores consistently in the 0.93-0.95 range,** making it highly reliable for clinical applications.

- The adversarial training approach successfully balances pixel-wise accuracy with perceptual quality, though the more complex training dynamics require careful hyperparameter tuning.

- **Turbo U-Net** offers computational efficiency with only **8M parameters** while maintaining reasonable performance **(SSIM 0.8118)**, making it suitable for resource-constrained environments.

- **Patient-level splitting** proved crucial for all models, ensuring performance metrics reflect true generalization capability rather than patient-specific overfitting. The consistent performance across validation and test sets validates this approach.

● **Error patterns** suggest occasional confusion in regions with similar tissue intensities, but all models maintain clinically acceptable accuracy for radiotherapy applications with MAE values well below 100 HU threshold.[2][3][1]

8. Conclusion

● **U-Net with Local Decoder** is the recommended model given the highest **SSIM (0.8725)** and superior anatomical detail preservation.

● For deployment under computational constraints, Turbo U-Net offers an excellent accuracy-efficiency trade-off with 8M parameters.

● **All three models** demonstrate clinical viability for MRI-only radiotherapy workflows with MAE values suitable for dose calculation accuracy.

● **Patient-level data splitting** establishes a new standard for medical imaging AI evaluation, preventing data leakage and ensuring valid performance assessment.

● **Potential future improvements include**:

(1) Clinical validation studies in real radiotherapy workflows

(2) Extension to other anatomical regions and imaging protocols

(3) Integration with treatment planning systems for dosimetric validation

(4) Multi-center dataset validation for broader generalizability

(5) Investigation of ensemble methods for improved robustness

● This work represents a significant advancement toward MRI-only radiotherapy planning, potentially reducing patient imaging burden while maintaining treatment quality standards.

Future Work

To build upon these promising results and advance toward clinical integration, the following research directions are proposed:

1. **Clinical Validation Studies:** Conduct rigorous clinical trials to assess the performance of the generated sCTs in real-world radiotherapy workflows, including qualitative evaluation by radiation oncologists and medical physicists.
2. **Extension to Other Anatomies:** Adapt and validate the models for other anatomical regions, such as the pelvis or abdomen, to broaden the applicability of MRI-only workflows where they would also be highly beneficial.
3. **Dosimetric Validation:** Integrate the best-performing models with treatment planning systems for direct dosimetric validation, to confirm that the observed low MAE values translate into clinically acceptable dose distributions.
4. **Multi-Center Dataset Validation:** Validate the models on large, diverse datasets from multiple institutions to ensure their generalizability across different scanner manufacturers, field strengths, and imaging protocols.
5. **Investigation of Ensemble Methods:** Explore the use of ensemble techniques, which combine predictions from multiple models, to potentially enhance predictive robustness and mitigate outlier errors.

This work contributes to a future where the burden of medical imaging on patients can be significantly reduced while maintaining, and potentially improving, the high standards of quality and precision required in modern radiotherapy.

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